LINEAR ALGEBRA

CHAPTER : 5

VECTOR SPACES

5.1: Real Vector Spaces:

Introduction:

You have studied about vectors and its properties in a 2-space or 3-space. i.e, when vectors were represented as doublets or point form (x, y) and triplets (x, y, z) or in a matrix form $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

We know that, the transpose of a matrix exchanges its columns with rows.

Ex: If $A = \begin{bmatrix} 1 & 2 & 6 \\ 5 & 3 & 8 \\ 9 & 8 & 7 \end{bmatrix}$, then $A^T = \begin{bmatrix} 1 & 5 & 9 \\ 2 & 3 & 8 \\ 6 & 8 & 7 \end{bmatrix}$

A matrix does not have to be square for its transpose to exist. For ex: if $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \end{bmatrix}$, then $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 4 & 6 \end{bmatrix}$

The rules of matrix transposition:

- The transpose of $A+B$ is $(A+B)^T = A^T + B^T$
- The transpose of $A \cdot B$ is $(A \cdot B)^T = B^T \cdot A^T$
- The transpose of $A \cdot B \cdot C$ is $(A \cdot B \cdot C)^T = C^T \cdot B^T \cdot A^T$
- The transpose of A^{-1} is $(A^{-1})^T = (A^T)^{-1}$.

A symmetric matrix has its transpose equal to itself i.e $A^T = A$.

Eg: $A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix}$

Note:

→ If you have a matrix X. that is not symmetric and you multiply it with its transpose X^T, say $X..X^T$ you get a symmetric matrix.

Ex: $X = \begin{bmatrix} 1 & 4 \\ 3 & 5 \\ 2 & 1 \end{bmatrix}$, then $XX^T = \begin{bmatrix} 1 & 4 \\ 3 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 23 & 6 \\ 23 & 34 & 11 \\ 6 & 11 & 5 \end{bmatrix}$

Proof: A matrix is symmetric iff $(A^T) = A$.

Now consider, $(X.X^T)^T = (X^T)^T . X^T = X.X^T$ (equal).

→ Inverse of a symmetric matrix is also symmetric.

Linear algebra is in essence a study of vector spaces, and this study of vector spaces is primarily devoted to finite dimensional vector spaces.

We start with an example,

Example: 1. (Vector space R^2 — all 2-dimensional vectors).

The vectors are of the form $xi+yj$ (are 2-D vectors) or (x,y) is $(1,1)$, $(2,4)$, $(3,2)$ etc. (we can prove this algebraically also) They represent the xy-plane.

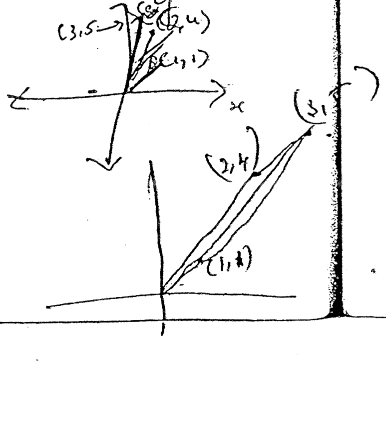
what makes these vectors vector spaces is, you add any two vectors $\vec{u}+\vec{v}$, the resultant is also a vector in xy-plane.

Say $(1,1) + (2,4) = (3,5) \in R^2$.

and you multiply any vector by a scalar k, say $k\vec{u} \in V$.

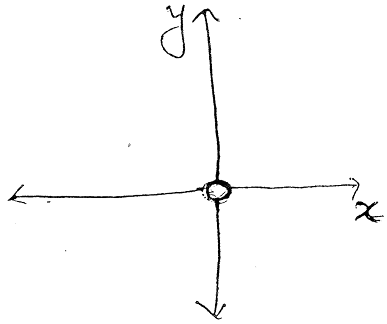
Ex: $(-1)(1,1) = (-1,-1) \in R^2$

$(0)[2,4] = [0,0] \in R^2$



There are other 8 axioms that the vectors must satisfy for them to make a space.

Does that mean all real spaces are vector spaces?
No, Say we consider the vectors in $R^2 - \{0,0\}$



Now we verify the condns.

$(1,1) + (2,1) = (3,2) \in R^2$

$(1,1) + (-1,-1) = (0,0) \notin R^2$

we see with multiplication

$(-1)\,(-3,4) = (-3,4) \in R^2$

But $0\,(3,4) = (0,0) \notin R^2$.

∴ The vectors doesnot form a Vector space.

|||ly only 1st quadrant +ve vectors are considered.
[It forms a subspace we ~~come~~ discuss later].

|||ly for

Ex-2: Vector space R^3 – all vectors with 3-components (all 3-D vectors).

Ex-3: Vector space – R^n – all vectors with n-components.

ie $\vec{V} = \{ v_1, v_2, \ldots, v_n \}$.

Addition in R^n:

$\vec{u} + \vec{v} = (u_1 + v_1,\ u_2 + v_2,\ u_3 + v_3,\ \ldots,\ u_n + v_n)$

Scalar multiplication in R^n:

$\alpha u = (\alpha u_1, \alpha u_2, \ldots, \alpha u_n)$, where '$\alpha$' is a scalar.

— x —

→ Now we give you the formal definition for Vector space.

Defn: Vector Space :-

Let V be an arbitrary set of objects ~~or~~ on which two operations are defined, addition and multiplication by scalars or scalar x^n

If the following axioms are satisfied for any scalars 'a' and 'b' and all objects u, v, w in V, then we call V a vector space and the objects ~~as~~ in V as vectors.

01) If u and v are objects in V, then $u+v \in V$ (Closed)

02) $u+v = v+u$ (Commutative)

03) $u+(v+w) = (u+v)+w$ (Associative)

04) There is an objects '0' in V, called a zero vector for V, such that $0+u = u+0 = u$ zero vector $\forall u \in V$

Identity (Existence of 0 as an additive identity)

05) For each u in V, there is an object '$-u$' in V called a negative of u, such that $u+(-u) = (-u)+u = 0$. Identity

(Existence of additive inverse)

06) $au \in V$ (closed x^n), where 'a' is a scalar.

07) $a(u+v) = au+av$ (left distributive) (scalars distribute over vector addition)

08) $(a+b)u = au+bu$ (right distributive)

09) $a(bu) = (ab)(u)$ Associative law for multiplication by scalars

10) $1.u = u$. (Identity) Number 1 is a multiplicative identity

(vi) If a is any scalar and u is any object in V, then au is in V.

Note:

→ The def^n of vector space specifies neither the nature of the vectors nor the operations.

→ Any object can be a vector, and the operations of $\oplus^n$ and scalar multiplication may not have any relationship to the standard vector operations in R^n.

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Scalars
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→ The only requirement is that the ten vector space axiom to be satisfied
~~Set to be a vector space is~~
~~that the ten vector properties is satisfied.~~
→ Scalars are real ⇒ real vector space, complex ⇒ Complex V.S

Example - 1 (2) in Text:

S.T the set V of all 2x2 matrices with real entries is a vector space if vector addition is defined to be matrix addn & vector scalar multiplication is defined to be matrix scalar multiplication.

Soln:-

Let the objects of V be, $u = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}$ & $v = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$.

$= u + 0 = u$
$\forall u \in V$
called
$u = 0$.
Identity

Axiom - 1:

S.T $u+v \in V$, i.e, we must show that $u+v$ is a 2x2 matrix (at various obj in V)

$$u+v = \begin{bmatrix} u_{11}+v_{11} & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \end{bmatrix}.$$ It follows from matrix addn

Axiom - 2:

$$u+v = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = v+u$$

lll^ly Axiom - 3: also follows

Axiom - 4:

S.T $0+u = u+0 = u$ for all $u \in V \in M_{2\times2}$.

ie we must find an object 0 in V

ie let $\vec{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$\therefore 0+u = u+0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \vec{u}.$$

the
tions of
y
R^2

Axiom-5:

we must show that each object 'u' in V has a negative '-u' such that $u+(-u)=0$ & $(-u)+u=0$.

This can be done defining the '-ve' of 'u' to be

$$-u = \begin{bmatrix} -u_{11} & -u_{12} \\ -u_{21} & -u_{22} \end{bmatrix}$$

$\therefore$ $u+(-u)=0$ III^{ly} $(-u)+u=0$.

Axiom-6:

$au \in V$

i's for any real scalar 'a', au should be a 2x2 matrix

$$au = \begin{bmatrix} au_{11} & au_{12} \\ au_{21} & au_{22} \end{bmatrix}$$ & consequently is an object in V.

Axiom- 7, 8, 9:

Are satisfied by the properties of matrix.

ie i) $a(B+C)=aB+aC$ ii) $(a+b)C = aC+bC$

iii) $a(bC) = (ab)C$.

Axiom- 10:

$$1.u = 1\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = u.$$

$\therefore$ Set of all 2x2 matrices is a Vector space under given operations.

Example-2:

Determine whether the set $V = R^2$ is a vector space or not, where the addition and scalar multiplication operations are defined as, if $u = (u_1, u_2)$ & $v = (v_1, v_2)$, then

$$u+v = (u_1+v_1, u_2+v_2) \quad \& \quad ku = (ku_1, 0)$$

where k is any scalar.

Solu^n:-

All the axioms will be satisfied, except for

Axiom-10

If $u = (u_1, u_2)$ we have to show that $1.u = u$

$$1.u = (1.u_1, 0) = (u_1, 0) \neq u.$$

Thus the given set $V = R^2$ is not a vector space with the stated operations.

Exercise - 5.1:

Determine which sets are vector spaces under the given operations. For those that are not, list all axioms that fail to hold.

① $V = R^3 = (x, y, z)$ ⇒ The set of all triples of real nos (x, y, z) with operations →

$$(x, y, z) + (x', y', z') = (x+x', y+y', z+z') \quad \&$$

$$k(x, y, z) = (kx, y, z)$$

Solu^n:-

Axiom - 8: fails

$$(k+l)u = ku + lu.$$

$$(ku + lu) = (kx, y, z) + (lx, y, z) = ((k+l)x, 2y, 2z)$$

& $(k+l)u = ((k+l)x, y, z)$

$\therefore (k+l)u \neq ku + lu$ ∴ V is not a vector space

02) The set of all triples of real numbers (x, y, z) with the operations

$(x, y, z) + (x', y', z') = (x+x', y+y', z+z')$

& $k(x, y, z) = (0, 0, 0)$

A-4
$0 = (0,0,0)$

Solu^n: Axiom-10 : Fails

$1.u = u$

A-5
$-u$

$1.u = 0$.

$(x,y,z)+(-u) = 0$
$(-x, -y, -z)$

Let $u = (x, y, z)$, then $1.u = (0, 0, 0) \neq u$.

∴ The given set of all triples is not a vector space with the stated operations.

03) The set of all pairs of real numbers (x, y) with the operations, $(x, y) + (x', y') = (x+x', y+y')$

& $k(x, y) = (2kx, 2ky)$

Solu^n: Axiom - 9 & 10 : fails

Axiom-9:

$k(lu) = (kl)(u)$

let $u = (x, y)$

$k(lu) = k[2lx, 2ly]$

$= (4lkx, 4lky)$

& $(kl)(u) = (2klx, 2kly)$

∴ $k(lu) \neq (kl)u$

Axiom-10:

$1.u = (2(1)x, 2(1)y)$

$= (2x, 2y)$

$\neq u$.

∴ The set of all pairs of real no's is not a vector space with the stated operations.

x, y, z)

04) The set of all real no's x with the standard operations of addn & multiplication,

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$u = x$
$v = y$

Soln:- It is a vector space.

tor space.

05) The set of all pairs of real numbers of the form $(x, 0)$ with the standard operations on R^2;

Soln: Axiom - 1:

$u = (x, 0)$, $v = (x', 0)$.

$u + v = (x + x', 0) \in V$

Axiom-2:

$u + v = v + u$

ith the

Axiom-3:

$(u+v)+w = u+(v+w)$.

Axiom-4: $(0,0) \in V$, such that

$\vec{0} + u = \vec{u} + 0 = u$. $= (x, 0)$

Axiom-5:

$-u = (-x, 0)$

$u + (-u) = (-u) + u = \vec{0}$.

, $a(1, y)$)

y)

Axiom-6:

$au \in V$

$a(x, 0) \in V$

Axiom- 7, 8 & 9 are satisfied

Axiom-10:

$1(x, 0) = (x, 0) = u \in V$

$\therefore$ The set of all pairs of real no's of the form $(x, 0)$ is a vector space.

vector

06) The set of all pairs of real no's of the form (x, y) where $x \geq 0$, with the standard operations on R^2.

Solu: Axiom-1: $(x, y) + (x', y') = (x+x', y+y') \in V$ $x, x' \geq 0$.

Axiom 5:

There is no object $(-u)$ in V such that $u+(-u) = (-u)+u = 0$.

Axiom 6:

$k(u) = (-1)(x, y) = (-x, -y) \notin V$.

$\therefore$ The given set is not a vector space.

07) The set of all n-tuples of real no's of the form $(x, x, ..., x)$ with the standard operations on R^n.

Solu: It is a vector space.

08) The set of all pairs of real no's (x, y) with the operations $(x, y)+(x', y') = \cancel{x+x', y+y'} \cdot (x+x'+1, y+y'+1)$ and

$k(x, y) = (kx, ky)$

Solu: Axiom-7:

$k(u+v) = ku+kv$

$k(u+v) = k[(x+x'+1), (y+y'+1)]$

$= [k(x+x'+1), k(y+y'+1)]$

$ku+kv = (kx, ky) + (kx', ky')$

$= (kx+kx'+1, ky+ky'+1)$

$\therefore k(u+v) \neq ku+kv$.

09

Solu

(x, y)
R^2

Axiom-8:

$(k+l)u = ku + lu$

$(k+l)u = ((k+l)x, (k+l)y)$

$ku + lu = (kx, ky) + (lx, ly)$

$= (kx + lx + 1, ky + ly + 1)$

$\therefore$ The given set is not a vector space with the stated operations.

form
n R?

Q9) The set of all 2x2 matrices of the form $\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

operations

Sol:

Axiom-1:

$u = \begin{bmatrix} a_1 & 1 \\ 1 & b_1 \end{bmatrix} \quad v = \begin{bmatrix} a_2 & 1 \\ 1 & b_2 \end{bmatrix}$.

then $u + v = \begin{bmatrix} a_1 + a_2 & 2 \\ 2 & b_1 + b_2 \end{bmatrix} \notin V$

Axiom-4:

* There is no object "0" such that $0 + \vec{u} = \vec{u} + 0 = \vec{u}$

Axiom-5:

If $-u = \begin{bmatrix} -a & -1 \\ -1 & -b \end{bmatrix}$

then $u + (-u) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin V$.

Axiom 6:

$ku \in V$

$0\begin{bmatrix} a & 1 \\ 1 & b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \notin V$.

$\therefore$ The given set is not a vector space.

10) The set of all 2x2 matrices of the form $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with matrix addition and scalar multiplication.

Solu: It is a vector space.

11) ~~The set of all real-valued functions 'f' defined~~

12) The set of all 2x2 matrices of the form $\begin{bmatrix} a & a+b \\ a+b & b \end{bmatrix}$ with matrix addition and scalar multiplication.

Solu:- Axiom-1:

$$u = \begin{bmatrix} a_1 & a_1+b_1 \\ a_1+b_1 & b_1 \end{bmatrix}, \quad v = \begin{bmatrix} a_2 & a_2+b_2 \\ a_2+b_2 & b_2 \end{bmatrix} \in V = M_{2\times 2}$$

$$u+v = \begin{bmatrix} a_1+a_2 & (a_1+a_2)+(b_1+b_2) \\ (a_1+a_2)+(b_1+b_2) & b_1+b_2 \end{bmatrix} \in V$$

Axiom-2:

~~$u+(v+w)$~~ $u+v = v+u$

Axiom-3: $u+(v+w) = (u+v)+w$.

Axiom-4:

choose a $0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in V$ such that

$u+0 = 0+u = u \in V$

Axiom-5:

$$-u = \begin{bmatrix} -a & -(a+b) \\ -(a+b) & -b \end{bmatrix}$$ such That

$u+(-u) = (-u)+u = \vec{0}$.

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c

Solu

Axiom – 6:

$ku \in V$

Axiom-7, 8, 9 are satisfied.

Axiom-10:

$1.u = \vec{u}$.

The given set is a vector space.

13. The set of all pairs of real no's of the form $(1, x)$ with the operations, $(1, y) + (1, y') = (1, y+y')$ and $k(1, y) = (1, ky)$.

Solu: Axiom-1:

$(1,y)+(1,y') = (1, y+y') \in V$

Axiom-2:

$u+v = v+u$

Axiom-3:

$u+(v+w) = (u+v)+w$.

Axiom-4:

Let zero vector be : $\vec{0} = (1, 0)$,

$0+u = (1,0)+(1,y) = (1,y) = u$

Axiom-5:

Let $-u = (1, -y)$, then $u+(-u) = (-u)+u = \vec{0}$.

$(1,y) + (-u) = \vec{0} = (1,0)$

$-u = (1,-y)$

Axiom-6:

'k' is any scalar say '-1', $(-1)(u) = (1, -y) \in V$

$0(u) = (1, 0) \in V$

Axiom-7:

$k(u+v) = ku + kv$

$k[(1, y+y')] = k(1,y) + k(1,y')$

$(1, k(y+y')) = [1, k(y+y')]$

Axiom-8:

$(k+l)u = ku + lu$

$(k+l)(1,y) = k(1,y) + l(1,y)$

$[1,(k+l)y] = [1,(k+l)y]$

Axiom-9: $k(lu) = (kl)u$ Axiom-10: $1u = u$

The given set is a vectorspace.

14. ~~The set of po~~

15. The set of all +ve real no's with operations, $x+y = xy$ and $kx = x^k$.

Solu: Let $x = u$ & $y = v$. For explanation check the xerox copy.

Axiom-1:

$u+v = x+y = xy \in V$

Axiom-2:

$u+v = xy = v+u \in V$

Axiom-3:

$u+(v+w) = (u+v)+w = xyz \in V$

Axiom-4:

Let the zero vector be equal to $\vec{0} = 1$ such that

$\vec{0} + u = 1 + x = 1 \cdot x = x = u$

III'ly $u + \vec{0} = u$.

Axiom-5:

$u = x$ & $-(u) = -(x) = x^{-1}$,

such that $u + (-u) = x + \frac{1}{x} = x \times \frac{1}{x} = 1$.

$(-u) + u = 1$.

$u + (-u) = 1$

$x + (-u) = 1$

Axiom 6:

If k is any scalar & u is any object in V then $ku \in V$

$0 \cdot x = x^0 = 1 \in V$.

$(-1)\, x = x^{-1} = \frac{1}{x} \in V$

$(-2)\, x = x^{-2} = \frac{1}{x^2} \in V$.

Axiom-7:

$k(u+v) = ku + kv$.

$k(u+v) = k(x+y) = k(xy) = (xy)^k$.

$ku + kv = kx + ky = x^k + y^k = x^k y^k = (xy)^k$.

Axiom-8:

$(k+l)\,u = ku + lu$

$(k+l)u = (k+l)\,x = x^{k+l}$

$ku + lu = kx + lx = x^k + x^l = x^{k+l}$

Axiom-9: is also satisfied

Axiom-10:

$1 \cdot u = 1 \cdot x = x^1 = x = u$.

∴ The given set is a vectorspace under ~~given~~ defined opes.

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16) The set of all pairs of real nos (x, y) with operations $(x,y)+(x',y') = (xx', yy')$ and $k(x,y) = (kx, ky)$.

Solu: Let $u = (x,y)$ & $v = (x',y')$.

Axiom-1:

$u+v = (x,y)+(x',y') = (xx', yy') \in V$

Axiom-2:

$u+v = v+u$

Axiom-3:

$u+(v+w) = (u+v)+w$

Axiom-4:

Let us choose the zero vector $\vec{0} = (1,1)$ such that $\vec{0}+u = (1,1)+(x,y) = (x,y) = u$

lly $u+\vec{0} = u$.

Axiom-5:

let $(-u) = -(x,y) = (-x,-y)$, such that

$u+(-u) = (x,y)+(-x,-y) = (-x^2, -y^2) \neq \vec{0}$.

Axiom-5 fails.

Axiom-6:

$ku \in V$

Axiom-7:

~~$(k+l)u = ku+lu$~~

$k(u+v) = ku+kv$

$k(u+v) = k(xx', yy') = (kxx', kyy')$

$ku+kv = (kx, ky)+(kx', ky') = (k^2xx', k^2yy')$

$\neq$.

$\therefore$ The ~~given~~ given set is not a vectorspace under defined operations

$\subseteq$

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dou ||.

$\subseteq$.

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Ex-4: Vector space of real valued functions :

Let V the set of all real valued functions defined on the entire real line $(-\infty, \infty)$.

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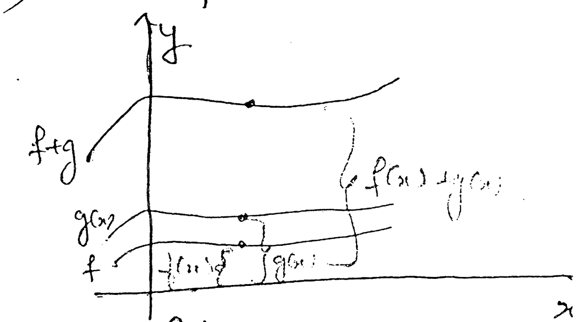
If $f(x), g(x) \in V$ & 'k' is any real no, we define the sum function $(f+g)(x) = f(x) + g(x)$ and Scalar multiple $(kf)(x) = kf(x)$

This vector space is denoted by $F(-\infty, \infty)$



Note:

→ $\vec{f} = \vec{g}$ is equivalent to $f(x) = g(x)$ $\forall x \in (-\infty, \infty)$

→ The vector $\vec{0}$ = constant function $= 0$ $\forall x$

→ $-\vec{f} = -f(x)$ is the reflection or inverse of $\vec{f}$.

Ex-5.1:

11. The set of all real-valued functions f defined everywhere on the real line and such that $f(1) = 0$, with the operations defined as $(f+g)(x) = f(x) + g(x)$ and

$$(kf)(x) = k f(x).$$

Solu: $u = f = f(x)$ & $v = g = g(x)$.

Axiom-1:

$u+v = (f+g)(x) = f(x) + g(x) \in V$ $\quad (f+g)(1) = f(1) + g(1) = 0 + 0 = 0$

Axiom-2:

$u+v = (f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x).$

Axiom-3:

$u+(v+w)=(u+v)+w$

Axiom-4:

Let zero vector function be of the form,

$\vec{0}=\theta(x)=0$

$\therefore (f+\theta)(x)=f(x)+\theta(x)=f(x)+0=f(x)$

Iden.

$f(x)=$

Soln:

Axiom-5:

$(f+(-f))x=f(x)+(-f(x))$

$=f(x)-f(x)=0=\theta(x)$

$[(-f)+f]x=0=\theta(x).$

Axiom-7:

$k[u+v]=ku+kv.$

$k[(f+g)(x)]=kf(x)+kg(x).$

~~$k[f+g(x)]$~~

Axiom-6:

$(kf)(x)=kf(x)\in V.$ $(kf)(1)=kf(1)=k0=0\in V$

Axiom-9:

$k(lu)=k(lf(x))$

$=kl\,f(x)$

Axiom-8:

$(k+l)u=(k+l)(f(x)).$

$=(k)f(x)+lf(x).\in V$

Axiom-10:

$(1.f)(x)=f(x)$

$\therefore$ The given set is a vector space under defined operations.

14. The set of polynomials of the form $a+bx$ with the operations

$$(a_0+a_1x)+(b_0+b_1x) = (a_0+b_0)+(a_1+b_1)x.$$

and $$k(a_0+a_1x) = ka_0+(ka_1)x.$$

(Its a vector space under given operation).

Solu:

Axiom 1: $f(x)=a_0+a_1x$; $g(x)=b_0+b_1x$

$(f+g)x = f(x)+g(x) = (a_0+a_1x)+(b_0+b_1x) = (a_0+b_0)+(a_1+b_1)x$

Axiom 2: $(f+g)x = (g+f)x$.

Axiom 3: $[f+(g+h)]x = [(f+g)+h]x$

Axiom 4: There is an object 0 in V such that $0+u = u+0 = u$

Let us choose $\theta = \theta(x) = 0+0x$

$(\theta+f)x = \theta(x)+f(x) = (0+0x)+(a_0+a_1x) = a_0+a_1x = f(x)$

Axiom 5: There an object $-u$ in V such that $f(x)+(-f(x)) = (-f(x))+f(x) = 0$

let $-f(x) = -a_0-a_1x$

$f(x)+-f(x) = (a_0+a_1x)+(-a_0-a_1x) = 0$.

Axiom 6: $\alpha f(x) = \alpha(a_0+a_1x) = (\alpha a_0+\alpha a_1x) \in V$.

Axiom 7: $\alpha[f(x)+g(x)] = \alpha[(a_0+b_0)+(a_1+b_1x)] = \alpha(a_0+b_0)+\alpha(a_1+b_1)x$

$\alpha f(x)+\alpha g(x) = \alpha(a_0+a_1x)+\alpha(b_0+b_1x) = \alpha(a_0+b_0)+\alpha(a_1+b_1)x$

Axiom 8: $(\alpha+\beta)f(x) = (\alpha+\beta)(a_0+a_1x) = (\alpha+\beta)a_0+(\alpha+\beta)a_1x$

$\alpha f(x)+\beta f(x) = \alpha(a_0+a_1x)+\beta(a_0+a_1x) = (\alpha+\beta)a_0+(\alpha+\beta)a_1x$

Axiom 9: $\alpha(\beta f(x)) = (\alpha\beta)f(x)$

$\alpha(\beta f(x)) = \alpha(\beta a_0+\beta a_1x) = \alpha\beta a_0+\alpha\beta a_1x$

$(\alpha\beta)f(x) = \alpha\beta(a_0+a_1x) = \alpha\beta a_0+\alpha\beta a_1x$.

Axiom 10: $1.f(x) = f(x)$

$1.f(x) = 1(a_0+a_1x) = a_0+a_1x = f(x)$

All the axioms are satisfied

$\therefore$ the set of polynomials of the form $a+bx$ is a

$f(x)$

$k0 = 0 \in V$

g:

$=k(lf(x))$

$kl\,f(x)$

defined

18) Prove that a line passing through the origin in R^3 is a vector space under the standard operations on R^3 / R^2. (we can do it for R^3 also)

Solⁿ: The set V is all the points that are on some line through the origin in R^2.

We know that the line must have the equation,

$ax+by=0$, for some 'a' & 'b', at least one not zero.

Also note that 'a' and 'b' are fixed constants, and aren't allowed to change (∵ those are always on the same line.

A point (x_1, y_1) will be on the line, and hence in V, provided $ax_1+by_1=0$.

We will show that V is closed under addition and scalar multiplication.

Let $u=(x_1, y_1)$ and $v=(x_2, y_2) \in V$

Axiom-1:

$u+v=(x_1+x_2, y_1+y_2) \in V$

now this point should satisfy the eqⁿ of line,

$$\therefore a(x_1+x_2)+b(y_1+y_2)$$

$$=(ax_1+by_1)+(ax_2+by_2)=0+0=0 \; (\because u, v \in V)$$

Axiom-6:

T.P $cu=(cx_1, cy_1) \in V$

$$a(cx_1)+b(cy_1)=c[ax_1+by_1]=c\cdot 0=0$$

Axioum-4:

$\vec{0}=(0,0)$ such that $u+0=u$.

Axiom-5:

$$-u=-(x_1, y_1)=(-x_1, -y_1) \qquad u+(-u)=(-u)+u=\vec{0}.$$

Verify the remaining axioms.

∴ The given set is a vector space.

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$v \in V$)

$+u = \vec{0}$.

Example-6:

Q. S.T Every plane through the origin in R^3 is a vector space.

Solu: Let V be any plane through the origin in R^3. We have to show that the points in V form a vector space under the standard addition and scalar multiplication operations for vectors in R^3.

Since R^n is a vector space under these operations, for $n=3$ i.e R^3 is itself a vector space under these operations.

Thus Axioms 2, 3, 7, 8, 9 and 10 hold for all points in R^3 and consequently for the points in the plane V.

∴ we need to show that axioms 1, 4, 5 and 6 are satisfied.

Since the plane V passes through the origin, it has an equation of the form

$$ax+by+cz=0 \longrightarrow (1)$$

eqn of a line in 3-D is $\frac{x-x_0}{m} = \frac{y-y_0}{n} = \frac{z-z_0}{l}$ which passes through the pt (x_1, y_1, z_1), m, n, l → direction cosines

∴ If $\vec{u} = (u_1, u_2, u_3)$ & $\vec{v} = (v_1, v_2, v_3)$ are points in V then $au_1+bu_2+cu_3=0$ & $av_1+bv_2+cv_3=0$.

∴ $\vec{u}+\vec{v} = a(u_1+v_1) + b(u_2+v_2) + c(u_3+v_3) = 0$.

∴ The co-ordinates of the point $\vec{u}+\vec{v} = (u_1+v_1, u_2+v_2, u_3+v_3)$ lies in the plane V.

∴ Axiom-1 is satisfied.

Axiom-4:

Let $\vec{0} = a(0)+b(0)+c(0) = 0$, $\vec{0}+\vec{u} = (0,0,0) + (au_1, bu_2, cu_3) = 0$

$\Rightarrow (0+au_1+0+bu_2+0+cu_3) = 0$.

$|||^{ly}$ $\vec{u}+\vec{0}$. $\Rightarrow au_1+bu_2+cu_3=0$

Axiom-6:

Let 'k' be any scalar, $k(\vec{u}) = aku_1 + bku_2 + cku_3 = 0$

$\Rightarrow (ku_1, ku_2, ku_3) \in V$

Axiom-5:

X^{ly} $\vec{u}$ with -1, we get $-au_1 - bu_2 - cu_3 = 0$.

$a(-u_1) + b(-u_2) + c(-u_3) = 0$.

Thus $-\vec{u} = (-u_1, -u_2, -u_3) \in V$.

Problem-19:

P.T the set of all points on a line that does not pass through the origin in R^2 is not a vector space under standard addition and scalar multiplication.

Solu:

The equation of the line will be, $ax + by = c$ where at least one of constants 'a' and 'b' is non-zero and the scalar $c \neq 0$.

Let $u = (x_1, y_1)$ and $v = (x_2, y_2) \in V$.

then P.T $u + v \in V$

$= (ax_1 + by_1) + (ax_2 + by_2)$

$= c + c$

$= 2c \neq c$.

$\Rightarrow u + v = (x_1 + x_2, y_1 + y_2) \notin V$.

So V is not closed under addition.

And also since $\vec{0} = (0,0) \notin V$.

So axiom-4 fails.

$\therefore$ The given set is not a vector space.

Ex-7:

$V = \{0\}$, $0+0=0$, $k0=0$, $\forall k$ is a vector space.

Theorem:

Let V be a vector space, 'u' a vector in V, and k a scalar, then

i) $0u = 0$ ii) $k0 = 0$ iii) $(-1)u = -u$

iv) If $ku = 0$, then $k = 0$ or $u = 0$.

5.2:

Subspaces:

Refer back to previous sections, Example - 1 and ~~example~~ Exercise problem - 18.

In Ex-1, we saw that R^2 is a vector space with the standard addition & scalar multiplication.

In Ex-P-18, we saw that the set of all points on a line through the origin in R^2 with the same operations was also a vector space.

What is so important about these two examples? Note, the set of all points in Ex-P-18 are also in the set of points in the vector space of Ex-1. But the opposite is not true, given a line we can find points in R^2 that aren't on the line.

What we have observed here is that, at least for some vector spaces, it is possible to take certain subsets of the original vector space and as long as we retain the definition of add^n & scalar multiplication we will get a new vector space.

It is possible for some sets to not be a new vector space.

Ex: In problem - 19, we've got a subset of R^2 with the standard add^n & scalar multiplication and yet it's not a vector space.

Note:

It is possible for one vector space to be contained within another vector space.

Definition: (Subspace) (atleast one element in it)

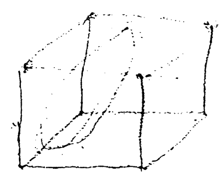
A subset of W of a vector space V is called a subspace of V if W is itself a vector space under the addition and scalar multiplication defined on V.

Theorem:

If W is a subset of vector space V, then W is a subspace of V iff

i) if $u, v \in W$, Then $u+v \in W$. (closed under addition)

ii) for any scalar k & $\forall\, u \in W$, then $ku \in W$. (closed under scalar multiplication)



→ 2 diff V.S one V.S can be contained within another V.S

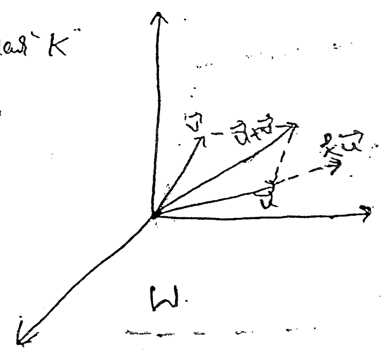
2) planes through the origin contained in 3D-space.

Example: 1- Subspaces of R^3:

Let W be any plane through the origin. Let $u, v \in W$. Then '$u+v$' must lie in W because it is the diagonal of the parallelogram determined by 'u' & 'v' (see fig)

& 'ku' must lie in W for any scalar 'k' because ku lies on a line through $\vec{u}$.

Thus, W is closed under addition & scalar multiplication, so W is a subspace of R^3.



Example-2: Lines through the origin are Subspaces:

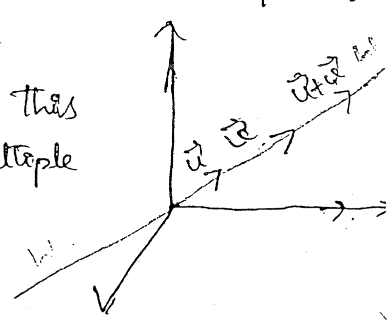
S.T a line through the origin of R^3 is a Subspace of R^3

Soln:- Geometrically:

Let W be a line through the origin in R^3.

It is evident geometrically that $\vec{u}+\vec{v}$ on this line also lies on the line and the scalar multiple of vector ($k\vec{u}$) on the line is on the line as well.

∴ It is a subspace of R^3.



Algebraically :

The parametric equation of a line passing through the origin in R^3 is given as,

$$W = at\,\hat{i} + bt\,\hat{j} + ct\,\hat{k}$$

let $\vec{u} = u_1 t\,\hat{i} + u_2 t\,\hat{j} + u_3 t\,\hat{k}$ & $v = v_1 t\,\hat{i} + v_2 t\,\hat{j} + v_3 t\,\hat{k} \in W$.

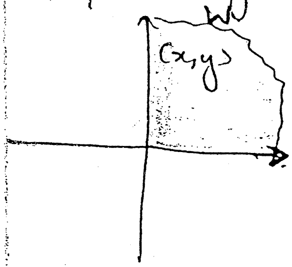
$$\vec{u}+\vec{v} = (u_1 t\hat{i} + u_2 t\hat{j} + u_3 t\hat{k}) + (v_1 t\hat{i} + v_2 t\hat{j} + v_3 t\hat{k})$$
$$= (u_1+v_1)t\,\hat{i} + (u_2+v_2)t\,\hat{j} + (u_3+v_3)t\,\hat{k}$$

$\therefore \vec{u}+\vec{v} \in W$.

$$k\vec{u} = k(u_1 t\hat{i} + u_2 t\hat{j} + u_3 t\hat{k})$$
$$= (ku_1)t\,\hat{i} + (ku_2)t\,\hat{j} + (ku_3)t\,\hat{k} \in W$$

$\therefore$ It is a subspace of R^3.

Example-3: Eg for not a Subspace of R^2:



Let W be the set of all points (x,y) in R^2 such that $x \geq 0$ & $y \geq 0$. These are the points in the first quadrant.

ie, $W = \{(x,y) / x \geq 0,\ y \geq 0\} = 1^{st}$ Q.

W is not a subspace of R^2 $\because$ it is not closed under scalar multiplication.

For ex: Let $\vec{v} = (2,2) \in W$. choose a scalar say $k = (-1)$,

$$k\vec{v} = -1(2,2) = (-2,-2) \notin W.$$

$\therefore$ Set W is not a subspace of R^2.

Note:

→ Every non-zero vector space V has at least two vector subspaces, vector space V itself as a subspace $\{0\}$ → zero subspace.

→ Subspaces of R^2
- $\{0\}$
- Lines through the origin
- R^2

Subspaces of R^3
- $\{0\}$
- Lines through the origin
- Planes through the origin
- R^3

These are the only subspaces of R^2 and R^3.

Eg:

Determine if the given set is a subspace of the given vector space.

a) Let W be the set of all points, (x,y), from R^2 in which $x \geq 0$. Is this a subspace of R^2?

Soln. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$ and

Since $x_1, x_2 \geq 0$, we also have $x_1 + x_2 \geq 0$,

$\therefore (x_1 + x_2, y_1 + y_2) \in W$. $\Rightarrow$ closed under addition.

$c(x,y) = (cx, cy)$ where 'c' is -ve scalar,

then $x > 0, c < 0$, we must have $cx < 0$,

$\therefore (cx, cy) \notin W \because cx < 0$.

$\therefore$ W is not a subspace of V.

b) Let W be the set of all points from R^3 of the form $(0, x_2, x_3)$. Is this a subspace of R^3?

$x + y = (0, x_2 + y_2, x_3 + y_3) \in W$.

$cx = (0, cx_2, cx_3) \in W$

$\Rightarrow$ W is a subspace of R^3.

c) Let W be the set of all points from R^3 of the form $(1, x_2, x_3)$. Is this a subspace of R^3?

$x + y = (2, x_2 + y_2, x_3 + y_3) \notin W$

$cx = (c, cx_2, cx_3) \notin W$

$\Rightarrow$ W is not a subspace of R^3.

Eg: Determine if the given set is a subspace of the given vector space.

a) Let W be the set of diagonal matrices of size $n\times n$. Is this a subspace of M_{nn}?

Solu: Let $u = \begin{bmatrix} u_1 & 0 & \cdots & 0 \\ 0 & u_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_n \end{bmatrix}$ & $v = \begin{bmatrix} v_1 & 0 & \cdots & 0 \\ 0 & v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_n \end{bmatrix}$

& 'c' be any scalar then,

$u+v = \begin{bmatrix} u_1+v_1 & 0 & \cdots & 0 \\ 0 & u_2+v_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_n+v_n \end{bmatrix}$ & $cu = \begin{bmatrix} cu_1 & 0 & \cdots & 0 \\ 0 & cu_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & cu_n \end{bmatrix}$

$\therefore$ W is a subspace of M_{nn}.

b) Let W be the set of matrices of the form $\begin{bmatrix} 0 & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$. Is this a subspace of M_{32}.

Solu: Let $u, v \in W$

$u+v = \begin{bmatrix} 0 & u_{12} \\ u_{21} & u_{22} \\ u_{31} & u_{32} \end{bmatrix} + \begin{bmatrix} 0 & v_{12} \\ v_{21} & v_{22} \\ v_{31} & v_{32} \end{bmatrix} = \begin{bmatrix} 0 & u_{12}+v_{12} \\ u_{21}+v_{21} & u_{22}+v_{22} \\ u_{31}+v_{31} & u_{32}+v_{32} \end{bmatrix} \in W$.

& $cu = \begin{bmatrix} 0 & cu_{12} \\ cu_{21} & cu_{22} \\ cu_{31} & cu_{32} \end{bmatrix} \in W$

$\therefore$ It is a subspace of M_{32}.

a

Solu

$C(-\infty,\infty)$

continuous derivatives of all orders is a

Note: in

c) b)

Solu

Note:

1) The set of $n \times n$ Symmetric matrices is a subspace of vector space M_{nn} of all $n \times n$ matrices. ($\because$ sum of two symmetric matrices is symmetric & a scalar multiple of a symm is ").

2) The set of $n \times n$ upper triangular matrices, the set of $n \times n$ lower triangular matrices and the set of $n \times n$ diagonal matrices all form subspaces of M_{nn}.

Example: Determine if the given set is a subspace of the given vector space.

a) Let $C[a,b]$ be the set of all continuous functions on the interval $[a,b]$. Is this a subspace of $F[a,b]$, the set of all real valued functions on the interval $[a,b]$.

Solu: A fact from Calculus, is that the sum of two continuous functions is continuous and multiplying a continuous function by a constants will give a new continuous function.

$C^{\infty}(-\infty,\infty)$ continuous derivatives of all orders is a

Note:- $[a,b)$ & (a,b) & $(-\infty,\infty)$ still the above result holds good.

$\therefore C[a,b]$ is a subspace of $F[a,b]$

Note: $|||^{ly}$ $C'(-\infty,\infty) = \{$continuous 1^{st} derivative funs on $(-\infty,\infty)\}$ is a subspace of $F(-\infty,\infty)$.

b) Let P_n be the set of all polynomials of degree 'n' or less. Is this a subspace of $F[a,b]$?

It is a subspace of $F[a,b]$

Solu: Let $u = a_n x^n + \dots + a_1 x + a_0$ &

$v = b_n x^n + \dots + b_1 x + b_0$, then

$$u+v = (a_n+b_n)x^n + \dots + (a_1+b_1)x + (a_0+b_0)$$

$$cu = (ca_n)x^n + \dots + (ca_1)x + (ca_0).$$

In both the cases the degree of the new polynomial is not greater than 'n'. Of course in the case of scalar multiplication it will remain deg(n), but with the sum, it is possible that some of

c) Let W be the set of all polynomials of degree exactly n. Is this a subspace of $F[a,b]$?

Solu: ~~let~~ $n=3$. $u = ax^3 + bx^2 + cx + d$.

$v = -ax^3 + px^2 + qx + t$.

where 'a' is not zero & other constants may & may not be zero.

$u+v = (b+p)x^2 + (c+q)x + (d+t)$.

the sum had degree '2' and so it is not in W.

∴ for $n=3$, W is not closed under addition.

So Generalizing it to 'n', we get the same result.

∴ W is not a subspace.

d) Let W be the set of all functions such that $f(6)=10$. Is this a subspace of $F[a,b]$ where we have $a \le 6 \le b$?

Solu: If we don't have $a \le 6 \le b$, then this problem makes no sense, so we will assume that $a \le 6 \le b$.

Let $f = f(x)$ & $g = g(x) \in W$ is $f(6)=10$ & $g(6)=10$.

$(f+g)(6) = f(6) + g(6) = 10 + 10 = 20 \ne 10$.

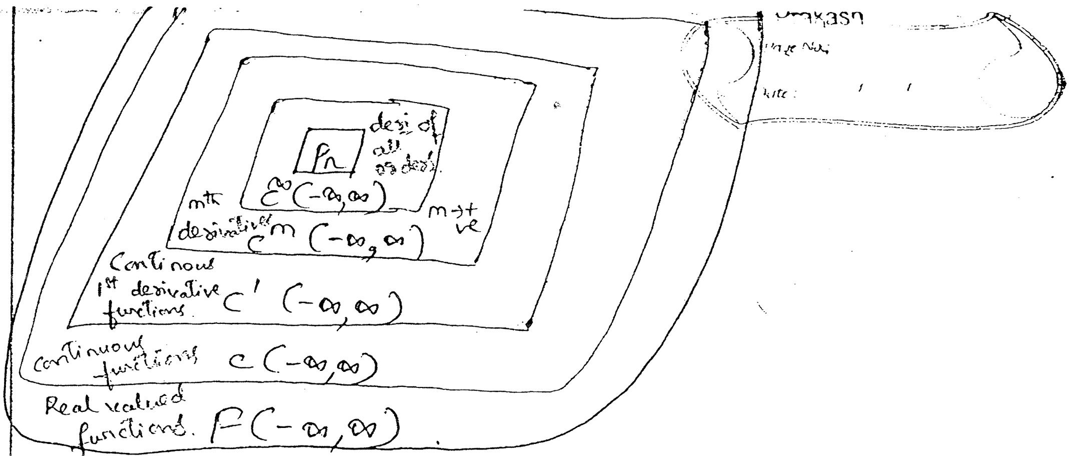
$(cf)(6) = cf(6) = c(10) \ne 10$.

∴ W is not closed under addition and scalar multiplication and so it is not a subspace.

Note:

~~Can~~ you ~~figure~~ out the value of ~~k~~, ~~given~~ ~~f(6)~~ ~~for~~ given ~~which~~ the ~~set W the set of all real~~ valued ~~functions~~. Given W is the set of all real valued functions for which $f(6)=k$ $a \le 6 \le b$? Are there any values of 'k' for which W will be a subspace of $F[a,b]$?

01)
a)
So
b)
c)
Solu
d)
Solu
don't do
2)
a)
b)



Exercise - 5.2:

01) Determine which of the following sets W are subspaces of R^3:

a) All vectors of the form $(a, 0, 0)$:

Solu: It is a subspace of R^3.

b) all vectors of the form $(a, 1, 1)$. $\Rightarrow$ No.

c) all vectors of the form (a, b, c) where $b = a + c$

Solu: Let $u = (a_1, b_1, c_1)$ where $b_1 = a_1 + c_1$ & $v = (a_2, b_2, c_2)$ where $b_2 = a_2 + c_2 \in W$

then $u + v = (a_1 + a_2,\ b_1 + b_2,\ c_1 + c_2)$ where

$b_1 + b_2 = (a_1 + c_1) + (a_2 + c_2)$

$= (a_1 + a_2) + (c_1 + c_2)$.

$\in W$

&

$cu = (ca_1, cb_1, cc_1)$ where $cb_1 = c(a_1 + c_1) \in W$

$\in W$

$\therefore$ It is a subspace of R^3.

d) all vectors of the form (a, b, c), where $b = a + c + 1$.

Solu: No.

don't do

2) Determine which of the following sets W are subspaces of M_{22}.

a) all 2×2 matrices with integer entries. $\Rightarrow$ No.
(not closed under scalar multiplication)

b) all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a + b + c + d = 0 \Rightarrow$ Yes.

c) All 2×2 matrices 'A' such that $\det(A)=0$.

Solu:

$|A+B| \neq |A| + |B|$

$\therefore$ W is not a subspace.

$\det(kA) = k^n \det(A)$
$\neq k \det(A)$

d) all matrices of the form $\begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$

Solu: Let $u+v = \begin{bmatrix} a_1 & b_1 \\ 0 & c_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ 0 & c_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ 0 & c_1+c_2 \end{bmatrix} \in W$

& $cu = \begin{bmatrix} ca_1 & b_1 \\ 0 & cc_1 \end{bmatrix} \in W$

$\therefore$ W is a subspace of M_{nn}.

3. Determine which of the following are subspaces of P_3.

a) All polynomials $a_0 + a_1x + a_2x^2 + a_3x^3$ for which $a_0=0$.

Solu:-
$u = a_0 + a_1x + a_2x^2 + a_3x^3$ &
$v = b_0 + b_1x + b_2x^2 + b_3x^3$ s.t $a_0=0$ & $b_0=0 \in W$

$u+v = \underbrace{(a_0+b_0)}_{c_0} + (a_1+b_1)x + (a_2+b_2)x^2 + (a_3+b_3)x^3$.

where $a_0+b_0=0$

and it is possible that some of the coefficients cancel out to zero ~~&~~ hence reduce the degree of the polynomial. ~~Both the both~~

$c(u) = (ca_0) + (ca_1)x + (ca_2)x^2 + (ca_3)x^3$, where $ca_0 = 0$

But in both the case the degree of the new polynomials is not greater than '3'.

$\therefore$ W is a subspace of P_3.

b).

c.

Solu:

e) a

a

Solu:

d) a

a

Solu:

b) all polynomials $a_0+a_1x+a_2x^2+a_3x^3$ for which $a_0+a_1+a_2+a_3=0$.

Solu: Let $u=a_0+a_1x+a_2x^2+a_3x^3$ & $v=b_0+b_1x+b_2x^2+b_3x^3$ $\in W$ s.t $a_0+a_1+a_2+a_3=0$ &

$b_0+b_1+b_2+b_3=0$

T.P $u+v\in W$:

$u+v=(a_0+b_0)+(a_1+b_1)x+(a_2+b_2)x^2+(a_3+b_3)x^3$,

s.t. $(a_0+b_0)+(a_1+b_1)+(a_2+b_2)+(a_3+b_3)$

$=(a_0+a_1+a_2+a_3)+(b_0+b_1+b_2+b_3)$

$=0+0=0$.

$\therefore$ $u+v\in W$.

T.P $cu\in W$

$cu=(ca_0)+(ca_1)x+(ca_2)x^2+(ca_3)x^3$,

s.t $ca_0+ca_1+ca_2+ca_3=c(a_0+a_1+a_2+a_3)$

$=c(0)$

$=0$.

$\therefore$ $cu\in W$.

$\therefore$ W is a subspace of P_3.

c) all polynomials $a_0+a_1x+a_2x^2+a_3x^3$ for which a_0, a_1, a_2 and a_3 are integers.

Solu: Not closed under scalar multiplication.

$\therefore$ W is not a subspace of P_3.

d) all polynomials of the form a_0+a_1x, where a_0 and a_1 are real no's..

Solu: W is a subspace of P_3.

4. Determine which of the following are subspaces of the space $F(-\infty, \infty)$.

a) all f such that $f(x) \leq 0$ for all 'x'.
(not closed under scalar multiplication)

Solu: W is not a subspace of $F(-\infty, \infty)$

b) all f such that $f(0) = 0$.

Solu: W is a subspace of $F(-\infty, \infty)$.

c) all 'f' such that $f(0) = 2$.

Solu: Not closed under addition and scalar multiplication.
W is not a subspace of $F(-\infty, \infty)$.

d) all constant functions.

Solu: W is a subspace of $F(-\infty, \infty)$.

e) all 'f' of the form $k_1 + k_2 \sin x$, where k_1 and k_2 are real no's.

Solu: Yes.

5. Determine which of the following are subspaces of M_{nn}.

a) all $n \times n$ matrices A such that $tr(A) = 0$. (Yes)

Solu. $tr(A) \rightarrow$ trace of A', is defined to be the sum of the entries on the main diagonal of 'A'. [$A \rightarrow$ squ matrix] $\begin{bmatrix} 0 & a & b \\ a & 0 & \cdots \\ b & -1 & \cdots \end{bmatrix}$

b) al

e) a
h

d) al
nx

? th

b) all $n \times n$ matrices A such that $A^T = -A$ (skew symmetric)

tion.

c) all $n \times n$ matrices such that the linear system $Ax = 0$ has only the trivial solution.

l K_2 are

M_{nn}.

es)

d) all $n \times n$ matrices A such that $AB = BA$ for a fixed $n \times n$ matrix B.

Solution Spaces of Homogeneous Systems :

Theorem: If $Ax=0$ is a homogeneous linear system of 'm' equations in 'n' unknowns, then the set of solution vectors is a subspace of R^n.

Note: Suppose 'A' is an $n \times m$ matrix. The null space of 'A' is the set of all x's in R^n such that $Ax=0$.

Eg-7: Solution spaces that are subspaces of R^3.

Determine whether the solution space of the system $Ax=0$ is a line through the origin, a plane through the origin or the origin only. If it is a plane, find an eqn for it; if it is a line, find parametric equations for it. Consider the linear systems:

(a)
$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 3 & -6 & 9 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(b)
$$\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ -2 & 4 & -6 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(c)
$$\begin{bmatrix} 1 & -2 & 3 \\ -3 & 7 & -8 \\ 4 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(d)
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Each of these systems has three unknowns, so the solutions form subspaces of R^3. Geometrically, this means that each solution space must be a line through the origin, a plane through the origin, the origin only or all of R^3.

Soln.

a) By applying Gauss-Elimination method,

$$x = 2s - 3t$$
$$y = s$$
$$z = t.$$

$$\Rightarrow \quad x = 2y - 3z \quad \Rightarrow \quad x - 2y + 3z = 0$$

$\therefore$ This is the equation of the plane through the origin with $\hat{n} = (1, -2, 3)$ as a normal vector.

b) The solutions are,

$x = -5t$, $y = -t$, $z = t$

which are parametric equations for the line through the origin parallel to the vector $\vec{v} = (-5, -1, 1)$.

c) The solution is $x = 0$, $y = 0$, $z = 0$, so the solution space is the origin only, ie, $\{0\}$.

d) The soluns are $x = r$, $y = s$, $z = t$, where r, s, t have arbitrary values, so the sol.on space is all of R^3.

Example: Determine the null space of each of the following matrices.

Defn: The solution space of the homogeneous system of equations $Ax = 0$, which is a subspace of R^n, is called the nullspace of A.

a) $A = \begin{bmatrix} 2 & 0 \\ -4 & 10 \end{bmatrix}$

Soln:- To find the null space of A

Solve the system of equations $Ax = 0$.

$$\begin{bmatrix} 2 & 0 \\ -4 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{aligned} 2x_1 &= 0 \\ -4x_1 + 10x_2 &= 0 \end{aligned}$$

$$\Rightarrow x_1 = 0 \ \& \ x_2 = 0$$

The solution consists of the single vector ~~0~~ and hence the null space of A is $\{0\}$.

b) $B = \begin{bmatrix} 1 & -7 \\ -3 & 21 \end{bmatrix}$

Soln

$$\begin{bmatrix} 1 & -7 \\ -3 & 21 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \left.\begin{aligned} x_1 - 7x_2 &= 0 \quad (1) \\ -3x_1 + 21x_2 &= 0 \end{aligned}\right\}$$ both are scalar multiple of each other

So (1) $\Rightarrow$ $x_1 = 7x_2$

let $x_2 = t$; $x_1 = 7t$. 't' is any real no.

∴ The null space of 'B' will be all of the points that

c) $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Solu: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Every vector $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ in R^2 will be a solution to this system.

Hence the null space of 'O' matrix is all of R^2,

Ex-5.2:

6) Q^n as of Ex-7.

a) $A = \begin{bmatrix} -1 & 1 & 1 \\ 3 & -1 & 0 \\ 2 & -4 & -5 \end{bmatrix}$

Solu: By Gauss elimination method,

$$A = \begin{bmatrix} -1 & 1 & 1 \\ 3 & -1 & 0 \\ 2 & -4 & -5 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 + 3R_1} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & -2 & -3 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + R_2} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x + y + z = 0$$

$$2y + 3z = 0.$$

$$y = \frac{-3}{2} z$$

$$x = y + z.$$

Let $z = $ 't', then $y = -3/2\, t$

$$x = y + z = t - 3/2\, t = \frac{-1}{2} t.$$

$\therefore$ Eqn of a line : $x = \frac{-1}{2} t$; $y = \frac{-3}{2} t$; $z = t$.

ition to

R^2,

b) $A = \begin{bmatrix} 1 & -2 & 3 \\ -3 & 6 & 9 \\ -2 & 4 & -6 \end{bmatrix}$.

Solu:-

Line ; $x = 2t$, $y = t$; $z = 0$.

c) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ d) $\begin{bmatrix} 1 & 2 & -6 \\ 1 & 4 & 4 \\ 3 & 10 & 6 \end{bmatrix}$ e) $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 4 \\ 3 & 1 & 11 \end{bmatrix}$

Solu: Origin $\Downarrow$ Origin $\Downarrow$ line : $x = -3t$, $y = -2t$, $z = t$.

f) $A = \begin{bmatrix} 1 & -3 & 1 \\ 2 & -6 & 2 \\ 3 & -9 & 3 \end{bmatrix} \Rightarrow$ plane ; $x - 3y + z = 0$.

Solu:-

$$\begin{bmatrix} 1 & -3 & 1 \\ 2 & -6 & 2 \\ 3 & -9 & 3 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & -3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x - 3y + z = 0.$$

]

Definition : Linear Combination

A vector W is called a linear combination of the vectors $v_1, v_2, \ldots, v_n$ if it can be expressed in the form $W = k_1 v_1 + k_2 v_2 + \cdots + k_r v_r$

or

$\cancel{Z} = c_1 v_1 + c_2 v_2 + \cdots + c_r v_r$.

where $k_1, k_2, \ldots, k_r$ are scalars.

Note:

If $r = 1$, then W is a linear combination of a single vector v_1. ie, $w = k_1 v_1$ (a scalar multiple of v_1).

Example-8: Vectors in R^3 are linear Combinations of i, j and k.

Every vector $\vec{v}=(a,b,c)$ in R^3 is expressible as a linear Combination of the standard basis vectors,

$i=(1,0,0)$, $j=(0,1,0)$ and $k=(0,0,1)$ since.

$$\vec{V}=(a,b,c)=a(1,0,0)+b(0,1,0)+c(0,0,1)$$
$$=ai+bj+ck.$$

Example-9: Checking a Linear Combination

Show that $\vec{w}=(9,2,7)$ is a linear combination of $\vec{u}$ & $\vec{v}$ and that $\vec{w}'=(4,-1,8)$ is not a linear combination of $\vec{u}$ & $\vec{v}$.
Given $\vec{u}=(1,2,-1)$ and $\vec{v}=(6,4,2)$ in R^3.

Soln:

Find c_1 and c_2 such that $\vec{w}=c_1\vec{u}+c_2\vec{v}$.

ie $(9,2,7)=c_1(1,2,-1)+c_2(6,4,2)$

$$\Rightarrow \quad \cancel{c_1}+6c_2=9\ ;\quad 2c_1+4c_2=2\ ;\quad -c_1+2c_2=7.$$
$$\cancel{-c_1}+2c_2=7$$
$$8c_2=16$$
$$\boxed{c_2=2} \quad\Rightarrow\quad c_1=9-12$$
$$\boxed{c_1=-3}$$

$\therefore\ \vec{w}=-3\vec{u}+2\vec{v}$.

IIIly find c_1 & c_2 such that $\vec{w}'=c_1\vec{u}+c_2\vec{v}$,

$$(4,-1,8)=c_1(1,2,-1)+c_2(6,4,2)$$
$$\cancel{c_1}+6c_2=4.\ ;\quad 2c_1+4c_2=-1\ ;\quad -c_1+2c_2=8.$$
$$\cancel{-c_1}+2c_2=8$$
$$8c_2=12$$
$$\boxed{c_2=\tfrac{3}{2}} \qquad c_1=4-6\times\tfrac{3}{2}$$
$$\boxed{c_1=-5} \longrightarrow (-10+6 \neq -4 \neq -1)$$

The system of eqns is inconsistent, So no such scalars exist.

$\Rightarrow\ \vec{w}'$ is not a linear Combination of $\vec{u}$ & $\vec{v}$.